

Closed-Form Solution of Markowitz-Mean-Variance Optimization

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Suppose we have a portfolio p consisting of 2 assets A and B , with respective weights w_A and w_B . The Variance of the portfolio p is given by:

$$\text{Var}(p) = \text{Var}(w_A A + w_B B) = \text{Cov}(w_A A + w_B B, w_A A + w_B B) \quad (1)$$

Thanks to the bilinearity of covariance, we can expand the above equation as follows :

$$\begin{aligned} \text{Var}(p) &= w_A^2 \text{Var}(A) + w_B^2 \text{Var}(B) + 2w_A w_B \text{Cov}(A, B) \\ &= w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}, \text{ where } \rho_{AB} \text{ is the correlation coefficient.} \end{aligned} \quad (2)$$

Thanks to this equation, we can see that indeed 2 assets with negative correlation would reduce the variance and therefore, the volatility squared of our portfolio.

Now, let's consider the case of n assets, with respective weights $\mathbf{w} = (w_1, w_2, \dots, w_n)$ and returns $r_1, r_2, \dots, r_n$. We thus have the return of the portfolio $r_p = \sum_{i=1}^n w_i r_i$. Treating r_i as random variables, we can deduce the following:

$$\text{Var}(r_p) = \text{Cov}(r_p, r_p) = \text{Cov}\left(\sum_{i=1}^n w_i r_i, \sum_{j=1}^n w_j r_j\right) = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j) \quad (3)$$

On the other hand, let Σ be a covariance matrix of the returns of the assets, with $\Sigma_{ij} = \text{Cov}(r_i, r_j)$. We therefore have $(\Sigma \mathbf{w})_i = \sum_{j=1}^n \Sigma_{ij} w_j$ and consequently

$$\mathbf{w}^\top \Sigma \mathbf{w} = \sum_{i=1}^n w_i (\Sigma \mathbf{w})_i = \sum_{i=1}^n w_i \sum_{j=1}^n \Sigma_{ij} w_j = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \text{Cov}(r_i, r_j) = \text{Var}(r_p) \quad (4)$$